1.

A loop structure repeats a block of code multiple times while a condition structure is true or until a condition is met.

2.

while loop Checks the condition before executing the code, so it might execute zero times.

do while loop Executes the code first and checks the condition afterward, so it always executes at least once.

3.

I used an input validation loop in my oddSum and Displaydigit applications as they helped the user input a non negative integer

4.

a)

An infinite loop is a loop that never ends because its stopping condition is never reached.

b)

A logic error, like using the wrong condition.

An update error, like forgetting to change the loop-control variable.

c)

Overflow occurs when a variable passes the max value that its data type can store, sometimes causing it to change around to another value.

5.

Answer: 60 times.

6.

Answer: Any integer value that's x that's less than 123.

7.

Counter: Keeps track of how many times something happens. It usually increases or decreases by a fixed amount.like Counting how many times a loop runs, or Counting how many students passed a test.

Accumulator: Keeps a running total by adding values together. Like adding up a list of numbers, or Calculating the total cost of items in a shopping cart.

Both use variables that are updated repeatedly inside a loop.

8.

```
int sum = 0;

for (int i = 3; i <= 10; i++) {
    sum = sum + i;
}

System.out.println(sum);
```

Output: 52

The loop starts at 3, adds each integer to sum, and stops after adding 10.

9.

if you how many times the loop needs to run.

if the loop must play at least once before its condition is checked.

Question 11

a)

Answer: 10

b)

"my "

c)

"my string."

d)

"MY STRING."

e)

"my string."